Part Three

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1 Algorithmic Design and Implementation Details

The objective function is Mean Squared Error (MSE), defined as $MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$.

Then, the gradient equation used for parameter updates for step i can be calculated as:

$$w^{i+1} = w^i - \alpha \frac{\delta MSE}{\delta m}$$

$$\frac{\delta MSE}{\delta m} = \frac{-2}{n}\sum_{i=1}^{n}(y_i - (mx_i + b))$$

Algorithm 1 Gradient Descent

procedure GRADIENT DESCENT(X, y, η, N)
  Initialize W^0, b^0
  for $i = 0$ to N **do**
    $\text{err} = f(X, W, b) - y$
    $W^{i+1} = W^i - \eta \cdot \frac{2}{|X|} X^\top \text{err}$
    $b^{i+1} = b^i - \eta \cdot 2\,\overline{\text{err}}$
  end for
  return W^N, b^N
end procedure

Our dependencies were:

ucimlrepo 0.0.7, pandas 3.0.5, matplotlib 3.11.2, numpy 2.5.3, scikit-learn 1.9.1, statsmodels 0.15.0, python 3.12

We designated rows 501–630 as the testing set (130 samples) and the remaining 900 samples as the training set, and split into a features matrix and output vector for each set. Model parameters were not updated based on the test dataset.

We set a list of learning rates for α = [1e-9, 1e-8, 1e-7], initial coefficient weights matrix to be all zeroes, initial bias to be zero, and iterations to be 1000. We stop the algorithm after i=1000 completes.

Then, loop over lrs and find the coefficient weights matrix, bias, along with mse and r2 over time as requested.

2 Feature Scaling and Gradient Descent Dynamics

3 Linear vs. Non-Linear Feature Transformations

4 Univariate vs. Multivariate Model Dyanamics

5 Statistical Inference and Regression Analysis

6 Code and Reproducibility Access